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Human Capital Theory

Implications for Educational Development in Belize and the Caribbean

LEROY ALMENDAREZ

Introduction: Human capital theory and education

EDUCATION IS A KEY TO DEVELOPMENT in every society. The twenty-first century paradigm is shifting towards the enhancement of knowledge as a priority, as states connect their higher educational systems much more closely to their various economic development strategies. An expansion of educational opportunities and access is believed to promote economic growth.

Education is an economic good because it is not easily obtainable and thus needs to be apportioned. It is regarded as both a consumer and a capital good, because it offers utility (satisfaction) to a consumer and also serves as an input to develop the human resources necessary for economic and social transformation. Human capital theory rests on the assumption that formal education is necessary to improve the productive capacity of a population. In short, human capital theorists argue that an educated population is a productive population. Human capital theory emphasises how education increases the productivity and efficiency of workers by increasing the level of cognitive stock of economically productive human capability, which is a product of innate abilities and investment in human beings. The provision of formal education is seen as an investment in human capital, which proponents of the theory have considered as equally or even more worthwhile than that of physical capital.¹ In the past, economic strength was largely believed to be dependent on tangible physical assets such as land, factories and equipment. Labour was a necessary component, but increases in the value of the business came from investment in capital equipment. Modern economists seem to concur that in the new global economy, education and health care are the key to improving human capital and ultimately increasing the economic outputs of the nation.²

The journalist Thomas Friedman, in his bestselling book, *The World Is Flat*, wrote extensively about the importance of education in the new global knowledge economy.³ His popular book has exposed millions of people to human capital theory. The term itself is not introduced in the book, but evidence as to why people and education (in other words, human capital) are vital to a nation's economic success is a recurring theme.

Noted economist Adam Smith, in his 1776 book *The Wealth of Nations*, formulated the basis of what was later to become the science of human capital.⁴ Over the next two centuries, two schools of thought were distinguished. The first school distinguished between acquired capacities that were classified as capital and human beings. The second school of thought claimed that human beings themselves were capital. In modern human capital theory, all human behaviour is based on the economic self-interest of individuals operating within freely competitive markets.

According to Joel B. Babalola, the rationality behind investment in human capital is based on three arguments:

1. The new generation must be given the appropriate parts of the knowledge which has already been accumulated by previous generations.
2. The new generation should be taught how existing knowledge should be used to develop new products, to introduce new processes and production methods and social services.
3. People must be encouraged to develop entirely new ideas, products, processes, and methods through creative approaches.⁵

A. Fagerlind and L.J. Saha posit that human capital theory provides a basic justification for large public expenditure on education in both developing and developed nations.⁶ The theory is consistent with the ideologies of democracy and liberal progression found in most western societies. Its appeal is based upon the presumed economic return of investment in education at both the macro and micro levels: efforts to promote investment in human capital are seen to result in rapid economic growth for society, while for individuals, such investment is seen to provide returns in the form of individual economic success and achievement. Human resources constitute the ultimate basis of the wealth of nations. Capital and natural resources are passive factors of production; human beings are the active agencies who accumulate capital,

exploit natural resources, build social, economic, and political organisations, and carry forward national development.

By applying the theory of human capital to educational systems, productivity is enhanced and sustained based on an increased and diversified labour force.⁷ George Psacharopoulos and Maureen Woodhall, suggest that economic appraisal of educational investment projects should take into account certain criteria:

- direct economic returns to investment, in terms of the balance between the opportunity costs of resources and the expected future benefits;
- indirect economic returns, in terms of external benefits affecting other members of society;
- the private demand for education and other factors determining individual demand for education;
- the geographical and social distribution of educational opportunities;
- the distribution of financial benefits and burdens of education.⁸

There seems to be a strong correlation between an educated population and technological innovation. The link is made explicit in what is termed 'investment in humans': workers need education in order to utilise new technologies, thereby increasing the total productivity and inducing economic growth. The accumulation of human capital through education and on-the-job training fosters economic growth by improving labour productivity, promoting technological innovation, and adaptation.

Empirical evidence of the human capital model

More than anything else, it has been the spectacular growth in East Asia that has given education and human capital their current popularity in the field of economic growth and development. Countries such as Hong Kong, South Korea, Singapore and Taiwan have achieved unprecedented rates of economic growth while making large investments in education. In the statistical analysis that accompanied its study, the World Bank found that improvement in education is a significant explanatory variable for East Asian economic growth.⁹ There are several ways of modelling how the huge expansion of education

accelerated economic growth and development. The first is to view education as an investment in human capital. A different view of the role of education in the economic success is that education has positive externalities: educate part of the community and the whole of it benefits.

The idea that education generates positive externalities is by no means new. Economists argue strongly for government's active support of education on the basis of its positive externalities – that society would gain from a more educated labour force and populace. Adam Smith reflected such progressive contemporary thought when he wrote that by educating its people, a society derives no inconsiderable advantage from their instruction. The more they are instructed, the less liable they are to the delusions of enthusiasm and superstition, which, among ignorant nations, frequently occasion the most dreadful disorders. Instructed and intelligent people are always more decent and orderly than ignorant ones. Smith viewed the externalities to education as important to the proper functioning not only of the economy but of a democratic society.¹⁰

Numerous studies have explored the extent to which educational attainment can contribute significantly to the production of overall output in an economy. Although macro studies have produced inconsistent and controversial results, several micro studies have shown a consistent positive relationship between education of the workforce and their labour productivity and earnings. The general finding is that individuals with more education tend to find better employment than those who are less educated. These findings provide a strong rationale for government and households to invest substantial portions of their resources in education, with the expectation that higher benefits will accrue over time. In that context, education is deemed an investment, equipping individuals with knowledge and skills that improve their employability and productive capacities, thereby leading to higher earnings in the future.¹¹ Studies have confirmed that human capital development is important as a development goal in its own right.¹²

Human capital development in the Caribbean

Human capital theory stresses the significance of education and training as the key to participation in the new global economy. The Organisation of

Economic Cooperation and Development (OECD), for example, claims that the radical changes to the public and private sectors of the economy introduced over recent years in response to globalisation will be severe and disturbing to many established values and procedures. The OECD believes that internationalism should be seen as an imperative in twenty-first century capitalism. This form of capitalism is based on investment in financial markets rather than in manufacturing of commodities, thus requiring dependence on electronic technology. The OECD also boldly asserts that internationalism is a means to improve the quality of education.¹³ In keeping with human capital theory, it has been argued that the overall economic performance of the OECD countries is increasingly based directly upon their knowledge stock and their learning capabilities. Clearly, the OECD is attempting to produce a new role for education in terms of human capital required in globalised institutions.

A decade ago, almost fifty-five million people of Latin America and the Caribbean were suffering from some sort of malnutrition. Eleven percent of the population was undernourished; almost 9 percent of children younger than five years suffered from acute malnutrition, and 19.4 percent from chronic malnutrition.¹⁴ Projections by the Economic Commission for Latin America and the Caribbean (ECLAC) and the International Labour Organization (ILO) show that in 2013 the unemployment rate is between 6.2 and 6.3 percent. This rate is only slightly lower than the reported rate of 6.4 percent in 2012. The employment rate was affected by a sluggish global economy, financial market volatility, and a reduction in the demand. The downturn in the region's economy resulted in a reduced demand for labour.¹⁵

The reality is that there continues to be a decline in the region's economic growth, with high unemployment rates, lower employee productivity, and high levels of poverty. CARICOM nations are being forced to implement necessary changes to compete more strategically in the global economy. Six pathways to economic development in the region have been identified: foreign trade; technological transfer; resource allocation; human capital formation; structural transfer; and savings and investment. Given the harsh reality and the lack of capital to invest, the focus of CARICOM has switched to increasing employee, organisational, national, and regional productivity through human capital formation, in order to enable CARICOM to be a player in the global economy and foster economic and social growth within the region.¹⁶ The

issue is determining the most effective ways to use the least resources to create human capital through human development strategies.¹⁷

In a comprehensive study of economic strategies of the more developed Caribbean nations conducted in 2003, it was highlighted that transformation of existing systems, policies and practices within CARICOM nations would be difficult to achieve if governments did not place more emphasis in their educational curricula on subjects that would lead to the emergence of greater economic growth. In an era where the Caribbean organisations are faced with changes generated by globalisation, there is an imperative need for human resources to be competent in participating in new modes of competition. Education is essential not only to enable people to share in the benefits of progress, but also to enable economies to ensure sustained development through competitiveness based on more intense knowledge use.¹⁸

There is a widening gap between the economies of the Caribbean and developed and emerging economies. In the Caribbean and Latin America, higher education is enduring a prolonged crisis where universities lack critical resources, technology and even intellectual capability to effectively prepare employees to compete in the global economy. The region has approximately 26 percent of its eligible individuals enrolled for post-secondary education.¹⁹ Many academicians and even politicians firmly believe that without a quality education, employees would not be able to produce at levels needed to compete in the global market and as such, there is an immediate need for focusing on education as a growth strategy. Compared with the curriculum in developed nations, there is also a critical need to focus on what is relevant in the business, cultural, political, and social environments. Business, political, and religious leaders also play a key role in ensuring that children stay in school. In many parts of the Caribbean, young children are spending all day working on farms, fishing, or just staying at home instead of going to school. These are the same individuals who will be hired by companies as employees and who are supposed to produce goods and services to supply customers around the world. It is plainly a system that fails to educate the population, producing a workforce that lacks the education, knowledge, and skills necessary to perform effectively. It is for these very reasons that education is one of the pillars of proactive social policy aimed at the implementation of the universal principles enshrined in the human rights declaration of the United Nations World Summit for Social Development.²⁰

Stock of human capital in the world

In a recent study, Robert J. Barro and Jong-Wha Lee used comparable data to estimate the current stock of human capital in the world; their focus was on the population aged fifteen years and over.²¹ The data set covers 146 countries during the period 1950–2010. Table 1 presents data in eight different regions and by gender.

Average years of schooling is used as a measure of human capital, because this can be determined for the entire workforce in most countries, it is fairly comparable across countries, and it is the most commonly used measure of human capital in research literature. Despite possible limitations, average years of schooling is still the most consistent and comparable country-level measure of human capital.

Table 1 shows that the average number of years of schooling in the world is 8.12 years, with males having 8.41 years of schooling and females, 7.84 years. A person in an industrialised country has the highest average length at 10.81 years, while a person in Latin America and the Caribbean has an average of 8.48 years (8.63 years for males and 8.33 years for females). There is a need for a strong policy thrust if those disparities in human capital are to be bridged.

Table 1: Average years of schooling by gender

Region	Male	Female	Average
Central Asia	9.35	9.99	9.69
East Asia and Pacific	8.47	8.01	8.24
Eastern Europe	10.24	9.95	10.09
Industrialised countries	10.92	10.71	10.81
Latin America and Caribbean	8.63	8.33	8.48
Middle East and North Africa	8.05	7.28	7.65
South Asia	6.41	4.79	5.62
Sub-Saharan Africa	5.98	4.89	5.43
World	8.41	7.84	8.12

Source: Robert J. Barro and Jong-Wha Lee, “A New Data Set of Educational Attainment in the World, 1950–2010”, NBER Working Paper No. 15902 (April 2010), 36, <http://www.nber.org/papers/w156902>.

Table 2: Annual growth rate in years of schooling, 1950–2010

Region	Male	Female	Average
Central Asia	1.24	1.67	1.46
East Asia and Pacific	1.38	2.28	1.76
Eastern Europe	1.05	1.40	1.23
Industrialised countries	.90	1.06	.98
Latin America and Caribbean	1.64	1.81	1.72
Middle East and North Africa	2.66	3.72	3.05
South Asia	2.00	2.75	2.26
Sub-Saharan Africa	2.67	3.18	2.89
World	1.53	1.88	1.69

Source: Barro and Lee, “New Data Set”, 7.

Table 2 shows that the annual growth rate in years of schooling for Latin America and the Caribbean was higher than the world annual growth rate and higher than the industrialised growth rate, but lower than that of the Middle East, North Africa, and South Asia.

Industrialised countries have much higher stocks of human capital than developing countries, so the next question is whether or not past performance in human capital accumulation indicates eventual convergence. Human capital convergence has been observed in past decades as developing countries continue to show an increase in growth rates.

Barro and Lee estimate that increasing average years of schooling by one year increases per capita GDP by 1.7–12.1 percent, depending on specification;²² D. Cohen and M. Soto calculate returns to years of schooling at 12.3–22.1 percent.²³ Testing the impacts of schooling quality on growth, it was found that a unit increase in a country’s average cognitive test scores increases per capita GDP growth rate by 1.2 to 2.0 percentage points. Moreover, increasing average math and science scores by one unit increases per capita GDP growth rates by 2.0 points, and by 2.3 points for low-income countries.²⁴ Overall, studies found that education significantly and positively correlated with economic growth and argued that causation runs from education and growth in line with human capital growth models.

Human capital development in Belize

In Belize, the plans of the government have continued to focus on human capital development, at least on paper, but statistics show that about 33 percent of primary school students are not entering high school. The implication of this is that they must be absorbed into the labour force of Belize, in some manner, by the provision of some skills training. The government's stated strategy is to focus on education development from pre-school to the tertiary level, but if 33 percent of primary-aged students are not matriculating to high school then it means that the numbers of educated persons in the population are cascading downward as the level of education increases. According to the figures from the annual budget, about 33 percent of the budget is allocated to the Ministry of Education, but if 92 percent of that amount is spent on salaries then there is only 8 percent available for infrastructure development. The three major challenges identified by the government are: (1) language difficulties, (2) teacher training, and (3) inadequate institutional support for teachers. Globally, there has been a paradigm shift in education to focus on science, technology, engineering and math, but in contrast the results of the Primary School Examination in math have been quite low.²⁵

Table 3 focuses on the human development index (HDI) for countries in the Caribbean and Central America. Comparatively speaking, Belize's HDI compares quite well with the three leading nations of CARICOM, that is, Jamaica, Trinidad and Tobago, and Barbados. Table 4 focuses on Belize and shows how life expectancy has improved since 1980. Life expectancy is one of the measures of human capital development, suggesting that people are living longer based on the level of health care, etc. The expected years of schooling increases gradually as well, but the mean years of schooling were higher between 1995 and 2000 and stabilised in the period 2005–2011.

In an effort to address some of the deficiencies and to create opportunities for school leavers or for those less academically inclined, the Ministry of Education of Belize recently took the initiative to offer training and development services through the Employment Training and Educational Services. The objective was to offer training to young people and adults that would give them the requisite employable skills or become entrepreneurs. This has been done through the establishment of the Training and Vocational Education System. Young people and adults can access a centre throughout the country.

Table 3: Human Development Index: Regional country comparisons

Country	Value	HD Level
Barbados	0.793	Very high
Mexico	0.77	High
Trinidad and Tobago	0.76	High
Jamaica	0.727	High
Belize	0.699	High
El Salvador	0.674	Medium
Honduras	0.625	Medium

Table 4: Human Development Index indicators for Belize 1980–2011

Year	Life expectancy at birth	Expected years of schooling	Mean years of schooling
1980	70.2	10.7	7.3
1985	71.5	10.7	7.6
1990	72.5	10.7	8
1995	73.1	10.7	8.5
2000	73.5	10.9	8.1
2005	74.5	12.6	7.7
2010	75.9	12.4	8
2011	76.1	12.4	8

Source: UNDP, Human Development Report, 2012.

The courses being offered include small engine repair, food preparation, catering, cake decorating, arts and crafts, cushion making, woodwork, computer training, and electricity; these are skills that are in great demand. Principals from the system can partner with the private sector to create opportunities for the trainees to be absorbed as soon as they are certified. In other instances,

alliances can be established with financial institutions so that start-up finance can be accessed on favourable terms.

As the effects and competitiveness are being felt in the global arena in which Belize operates, the strategy must be to ensure that the competitive skills are learned at the primary and secondary levels and matured through the tertiary level. The global marketplace is even more connected through wireless technology, and therefore, the education system must be cognisant and curricula must address this phenomenon with the urgency that is required. Tertiary-level institutions can partner with the business community and assess their needs so that the course offerings make the graduates of these institutions immediately marketable. This will provide a great impetus on satisfying the employment demands of the workforce. The teaching of science, technology, engineering and mathematics must be prioritised and students should be provided with a solid foundation so that they can undertake these areas of study and be able to compete in the global marketplace.

Conclusion

Since the purpose of a minimum standard of education should help to offset the effects of poverty, appropriate subsidies could in principle achieve the same result without compulsion. The effectiveness of voluntary investment in human capital is often underrated because subsidies to human capital usually cover, at best, only a portion of earnings forgone. If they covered all costs, including those forgone, almost all children would continue in school through to the age desired.

Developing countries are confronted by problems that could limit the capacity of expansion in education to stimulate growth and development: for example, underemployment, low absorptive capacity, shortage of professionals, regional imbalances, and brain drain. The persistence of many problems in spite of the various policy formulations and responses points to the need for a more focused, responsive, functional, and qualitative educational system. To contribute significantly to economic growth and development, education must be of high quality and also meet the skill-demand needs of the economy.

It is not a noble achievement for any sector of the economy to make only

a negligible contribution to economic growth, one which is commensurate neither with its lifespan nor with the investment that has gone into it. There is a need for more commitment by government not to interfere with decisions such as curriculum or teachers' responsibilities. Government, in its employment policies, should place more emphasis on specialisation and competence rather than paper qualification and ill-gotten certificates.

There are serious consequences for not investing in education and not meeting the development goal of education for all. They include: underdeveloped and underutilised human and social capital; loss of economic productivity; increased unemployment and underemployment; increased violence, crime, and risky behaviours; exploitation and marginalisation of youth, especially girls and young women; reduced social cohesion and disruption in civil society; and increased spending on remedial social welfare services and crime prevention. The architects of the Belizean education system must employ a vision that ensures the institutions of learning cater to the development of a functional workforce that can contribute to the economic development of Belize. 

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